

ProductionManagementAI — Product master — Basic Design Document (基本設計書)

Document control (改版履歴)

Field	Value
Document ID	004_BD
Category	UI and product reference data
System / subsystem	ProductionManagementAI / Product master
Work item	WI-006
Based on brief	WI-006 brief , design baseline 2026-09-29
Created / updated	Codex, 2026-09-29
Version	2 — reconciled with DEC-007–DEC-010 (unit-locking rule, measured quantities, mixed-unit impacts)

System overview

SCR-004 lets an **Admin** or **Operator** find, create, edit, and retire automobile parts without changing the database directly. Retiring a part leaves its historical orders readable. It is unavailable as a new product selection, while an old order may retain that product during an unrelated edit ([WI-006 decisions](#) DEC-001–DEC-004). To protect historical order integrity, a product's unit cannot change once referenced by any production order (DEC-010, REQ-057).

Requirement	Coverage
REQ-049	List, search, empty and error states
REQ-050–REQ-051	Create/edit form and validation
REQ-052, REQ-056	Retirement, SCR-001 picker, SCR-002 filter, historical orders
REQ-053	Seed and migration design in 004_DB
REQ-054–REQ-055	Role gate, Japanese UI, responsive and accessible behavior

Requirement	Coverage
REQ-057	Product unit locked on edit when referenced by orders

Overall configuration and architecture

The existing React app calls same-origin ASP.NET Core controllers backed by EF Core and PostgreSQL 17. `SCR-004` uses the shared authenticated header and Japanese message catalog. Product management uses new API operations while the existing `GET /api/products` remains available to production-order screens. No new external service or trust boundary is introduced; the [auth ADR](#) and the existing JSON-only mutation posture apply. The [004_DD-API](#) owns the exact contracts.

Function list

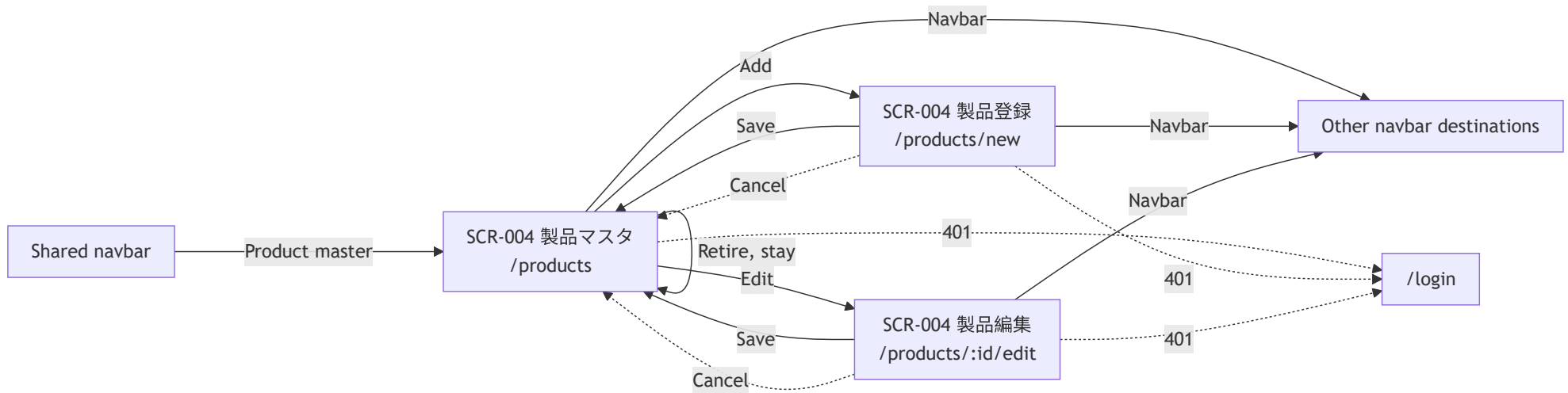
Function ID	Function	Requirements
FN-027	Search and page the product catalog	REQ-049
FN-028	Create an active product	REQ-050, REQ-054
FN-029	Edit mutable product fields	REQ-051, REQ-054
FN-030	Retire an active product	REQ-052, REQ-054
FN-031	Select a product on an order with retirement rules	REQ-052, REQ-056

Actors and business flow

1. An `Admin` or `Operator` opens 「製品マスタ」 (Product master) from the shared navbar, then searches by SKU or name and filters by active/retired state.
2. The maintainer chooses 「製品を登録」 (Add product), enters SKU, Japanese name, unit, and optional drawing number, then saves. A successful save returns to the catalog with the new row visible.
3. The maintainer chooses 「編集」 (Edit), changes name or drawing number (and unit if no orders reference this product), and saves. SKU is displayed but cannot be changed. If any production order references this product, unit selection is locked (disabled with an explanatory notice) to preserve the historical meaning of order quantities (REQ-057, DEC-010).
4. The maintainer chooses 「使用停止」 (Retire), reviews a confirmation naming the part, and confirms. The row remains in the catalog as retired.
5. On SCR-001, active parts are selectable for new or changed selections; a retired part already on an order remains visible and can be retained. SCR-002's product filter still includes retired parts to find historical orders.

Screen list and transition

Screen / mode	Name	Entry	Exit / next state
SCR-004 list	「製品マスタ」 (Product master), <code>/products</code>	Navbar, create/edit save or cancel	Create mode, edit mode, stay on list after retirement, navbar destinations, <code>/login</code> on 401
SCR-004 create	「製品登録」 (Add product), <code>/products/new</code>	List Add	List on save/cancel, <code>/login</code> on 401
SCR-004 edit	「製品編集」 (Edit product), <code>/products/:id/edit</code>	List Edit	List on save/cancel, <code>/login</code> on 401



The shared unsaved-change guard applies when leaving a dirty create/edit form, including navbar navigation. Retire confirmation is a dialog on the list, not a route. A 403 shows a forbidden state without attempting another write.

Screen design detail — SCR-004 Product master

0-1. Basic information

Item	Design
Routes	/products , /products/new , /products/:id/edit
API base	/api/product-master ; existing /api/products supplies order screens
Encoding	UTF-8
Error fallback	Local retry state for fetch failures; /login on 401; forbidden state on 403
Responsive	PC and SP layouts; list table becomes cards on SP
Authentication / authorization	Cookie session and Admin or Operator role, enforced by API
URL parameters	List query q , state , page ; edit route id is a UUID
Page title	Mode heading followed by ProductionManagementAI
Base/link/meta/style/script overrides	Not applicable — shared app document and bundled styles/scripts apply

1. Layout and mockup

1

ProductionManagementAI ダッシュボード 製造指示一覧 新規製造指示 製品マスタ

2

製品マスタ

製品を登録

3

製品コード・製品名で検索 状態：すべて 検索 クリア

4

製品コード	製品名	単位	図面番号	状態	操作
P-1001	ブレーキキャリパー	個	—	使用中 編集 使用停止	
P-1002	ブレーキディスクローター	枚	—	使用停止 編集	

5

30件 前へ 1 / 2 次へ 読み込み・エラー・結果通知

6

別モード：製品登録・編集フォーム 製品コード / 製品名 / 単位 / 図面番号

7

一覧上の確認ダイアログ：製品名 キャンセル / 使用停止

1

ProductionManagementAI メニュー

2

製品マスタ
[製品を登録]

3

製品コード・製品名で検索
状態: すべて
[検索] [クリア]

4

P-1001 ブレーキキャリパー
単位: 個 図面番号: — 使用中
[編集] [使用停止]

5

30件 前へ 1 / 2 次へ

6

別モード: 製品登録・編集
項目は1列に配置 保存 / キャンセル

7

確認ダイアログ: 製品名
キャンセル / 使用停止

No.	Region	Content and behavior
1	Shared header	Existing application name and navbar, with 「製品マスタ」 entry and current-page marker

No.	Region	Content and behavior
2	Heading/actions	List heading and 「製品を登録」 action; create/edit headings and save/cancel actions in form mode
3	Search/filter	SKU/name search, active-state filter, apply/clear actions; URL reflects applied values
4	Catalog	PC table or SP cards showing SKU, name, unit, drawing number, state, Edit, and Retire when active
5	Pagination/status	Total count, current page, previous/next; loading, empty, fetch error, and success notice
6	Product form	SKU (read-only in edit), name, fixed unit selection (locked/disabled with helper note in edit if referenced by orders), optional drawing number, field errors
7	Retire confirmation	Dialog naming the part with Cancel and Retire; focus returns to the invoking action

2. Content blocks and visible states

State	Visible result
Loading	Heading and a loading status; actions that depend on fetched data wait.
Empty catalog / no search match	Distinct Japanese messages; Add remains available.
Ready	Rows/cards and total; retired rows carry 「使用停止」 (Retired) text as well as styling.
Fetch failure	Error message and 「再試行」 (Retry), without stale data presented as current.
Validation failure	Inline errors, an error summary and focus on the first invalid field.
Concurrent edit / retirement	Explain that the record changed; offer reload without silent overwrite.
Save/retire success	Return to list or update the row; announce success politely.

3. Screen item definition

Item	Japanese UI text (English gloss)	Source / type	Rule
SKU	「製品コード」 (Product code)	Required text, maximum 50 characters	Trim ends on create, case-insensitive unique; immutable afterward
Name	「製品名」 (Product name)	Required text, maximum 200 characters	Japanese display name; whitespace-only invalid
Unit	「単位」 (Unit)	Required fixed selection	個, 本, 枚, 台, セット, kg, m. Mutable on creation; locked (disabled/read-only) in edit mode when any production order references this product (DEC-010, REQ-057)

Item	Japanese UI text (English gloss)	Source / type	Rule
Drawing number	「図面番号」 (Drawing number)	Optional text, proposed maximum 100 characters	Blank stored as null; exact length confirmed in DD
State	「状態」 (State)	Active / retired	No hard delete; retirement confirmed by user
Search	「製品コード・製品名で検索」 (Search by code or name)	Optional text	Applied only on Search, reset page to 1

4. Item value mapping

Value	Display
Active	「使用中」 (Active)
Retired	「使用停止」 (Retired)
All filter	「すべて」 (All)
Optional drawing number absent	「—」 (Not provided)

5. Validation checks

Check	User result	Requirement
Missing/blank SKU or name, missing unit	Explain at the field and block save	REQ-050–REQ-051
SKU duplicate after trim and case-insensitive comparison	Explain conflict; keep entered values	REQ-050
Unknown unit or too-long field	Explain allowed values/length; backend rejects independently	REQ-050–REQ-051
Attempted unit change on referenced product	Explain that unit cannot be changed because orders already reference this product; block save	REQ-057
Stale edit or retire version	Explain conflict and offer reload	REQ-051–REQ-052
Unauthorized action	Server returns 401/403; UI provides navigation or forbidden message	REQ-054

6. Item events

Event	Result
Apply/Clear filters	Update list URL and fetch page 1.
Add/Edit	Navigate to the create/edit mode; preserve list URL for return.

Event	Result
Save	Validate, submit once, navigate to list on success; keep form on error.
Cancel/navbar while dirty	Use existing unsaved-change confirmation before navigation.
Retire	Open named confirmation dialog; only confirmation submits; keep list row visible as retired.
401/403/API failure	Navigate to login, show forbidden state, or show retry/error respectively.

7. External identity linkage

Not applicable — existing cookie authentication and local `Admin / Operator` roles are reused; no external identity provider is added.

Impact on existing screens and data

`GET /api/products` must include active-state and unit information while continuing to supply SKU/name/ID to SCR-001 and SCR-002. - SCR-001: Filters new selections to active products, but includes its current retired product as a retained choice in edit mode. Displays the product's unit alongside order quantity. Supports measured decimal quantities (up to 3 decimal places) for `kg` and `m`, while enforcing positive integers for discrete units (REQ-058, DEC-008, DEC-009). The server allows an unchanged retired product ID on update and rejects a newly chosen retired ID. - SCR-002: Keeps retired products in its filter so historical orders remain findable. Displays unit alongside quantities in the table and details (REQ-059). - SCR-003: Dashboard aggregates order quantities only within the same unit; cross-unit summaries display order counts (REQ-060, DEC-007). Product ID and FK behavior remain unchanged; see [004_DB](#).

Non-functional design and open points

- **Accessibility:** keyboard-operable list, form and confirmation; visible focus; explicit labels and error text; no color-only state; PC/SP layouts and WCAG 2.2 AA. Locked unit input in edit mode is clearly identified to screen readers (e.g. `aria-disabled="true"` with helper text). The modern-web-guidance forms reference recommends semantic forms, visible labels, validation after interaction, and clear focus/error behavior.
- **Security:** `Admin / Operator` checks on every new API operation, including writes; UI gating alone is insufficient. Product fields contain no expected PII or secrets. Mutations remain JSON-only under the existing same-origin cookie/CSRF posture; revisit if CORS or hosting origins change.
- **Observability:** count list/create/update/retire outcomes without logging SKU, name, drawing number, or full bodies; detailed telemetry names belong in 004_DD-FN.
- **Migration:** additive columns and a per-part unit backfill for the 30 seeded rows; remove EF model-managed seed tracking without deleting the rows. The mapping and handling of any non-seed rows are specified in WI-006 DEC-005 and 004_DB.